

Complete Features & Technical Architecture Specification

A TikTok-Style Spaced Repetition Mobile Platform for Bacculaureate History & Geography

Version: 1.0.0 (Production)	Framework: Flutter 3.x / Dart 3.x	State & DI: Pure Riverpod 2.x
Architecture: Clean Architecture + Offline First	Supported Languages: Arabic, French, English	Target Audience: Bacculaureate Students

1. Executive Summary & Core Philosophy

Hafedh (هفدح) is a high-retention mobile learning app designed to solve the cognitive fatigue associated with traditional rote memorization in the Algerian Bacculaureate curriculum (History & Geography). By marrying the highly engaging, addictive micro-interaction pattern of vertical swipe feeds (TikTok style) with scientifically proven cognitive spaced repetition (SuperMemo SM-2 algorithm), Hafedh empowers students to master complex dates, personalities, definitions, and concepts in effortless, daily bite-sized sessions.

Key Architectural Pillars:

- **TikTok-Style Vertical Snapping:** Fast, addictive 60/120fps vertical feed designed for micro-learning.
- **6 Interactive Modalities:** QCM quizzes, 3D flip cards, date timelines, personality cards, definitions, memory advice.
- **3-Tier Priority Deck Re-Ordering:** Prioritizes unlearned and difficult cards with randomized internal ordering.
- **Offline-First Resilience:** Full offline caching, offline review queuing, and instant demo mode without server dependency.
- **Trilingual RTL/LTR Engine:** Complete localization in Arabic, French, and English with dynamic directionality.

2. Detailed Feature Breakdown

2.1 Ultra-Smooth TikTok-Style Vertical Feed

The core learning experience takes place in a full-screen vertical PageView.builder. Designed to run at continuous 60fps/120fps on modern mobile displays:

- **Allow Implicit Scrolling:** Pre-renders adjacent cards in memory, eliminating layout hitching and scroll stutters.
- **Zero-Rebuild Gesture Pipeline:** State listeners are isolated via Riverpod selectors, meaning swiping incurs zero tree rebuilds.
- **Repaint Boundaries:** Isolates GPU rasterization layers per card, preventing full-screen re-renders.
- **Auto-Advance Progression:** After submitting recall feedback, the feed smoothly animates to the next card after 650ms.

2.2 Six Distinct Interactive Card Modalities

To prevent cognitive monotony, the app renders 6 specialized card types dynamically matched to curriculum requirements:

Card Type	Visual Identity	Pedagogical Functionality & Interaction
1. QCM Quiz	Teal badge + 4 choice capsules	Interactive multiple-choice questions with instant green/red validation and detailed pedagogical explanations.
2. 3D Flip Card	Blue badge + 3D Matrix4 Flip	Realistic 3D flipping animation revealing standard Baccalaureate model answers and memory keys.
3. Historical Dates	Gold badge + Date highlight	Large prominent date numerals with interactive 'Reveal Historical Context' button and timeline clues.
4. Personality Card	Purple badge + Avatar halo	Biographical profile of key leaders/figures highlighting nationalities, key decisions, and historical roles.
5. Concept & Term	Emerald badge + Book icon	Glossary definition card highlighting core keywords and exam criteria for accurate memorization.
6. Memory Advice	Rose badge + Bulb icon	Mnemonic formulas, acronyms, and cognitive memory capsules designed to anchor hard-to-remember facts.

2.3 3-Tier Memorization Priority Deck Re-Ordering Algorithm

The deck dynamically arranges learning materials into 3 priority tiers based on the student's historical recall feedback:

- **Tier 1 (Top Priority):** Unreviewed cards + cards marked **0%** (■■■ ■■■■ / **Not Yet**). These appear first and are shuffled randomly for fresh variety.
- **Tier 2 (Middle Priority):** Cards marked **50%** (■■■ ■■■ / **Partially**). Placed in the middle queue for stabilization.
- **Tier 3 (Low Priority):** Cards marked **100%** (■■■■■■■■ / **Mastered**). Placed at the end of the session to avoid cluttering new material.
- **Real-Time Re-Insertion:** Marking a card 0% immediately re-inserts it 4 slots ahead in the active session for instant cognitive reinforcement.
- **Persistent Recall Memory:** All ratings are saved to local persistent storage (StorageService), maintaining the student's prioritized order across app restarts.

2.4 Comprehensive Multilingual Support (AR, FR, EN)

Full localization architecture powered by LanguageProvider and AppTranslations:

- **Arabic (■■■■■■■■ - ■■):** Default language with full Right-to-Left (RTL) layout and Tajawal typography.
- **French (Français - ■■):** Left-to-Right (LTR) layout with native French terminology.
- **English (English - ■■):** Left-to-Right (LTR) layout with modern educational phrasing.
- **Instant Switcher Touchpoints:** Available from the top feed header, user profile screen, and login screen.

2.5 Pixel-Perfect Responsiveness (`flutter\_screenutil\_plus`)

All visual components, paddings, icon bounds, and typography are mathematically scaled using flutter_screenutil_plus based on standard design viewport Size(390, 844) with .w, .h, .r, and .sp units. This guarantees zero visual clipping or overflows across all Android and iOS form factors.

2.6 Offline-First Architecture & Network Resilience

Engineered for students in areas with unreliable internet connectivity:

- **Instant Offline Demo Mode:** One-tap entry on the login screen allowing instant study with zero server connection.
- **Offline Review Queuing:** Offline card ratings are queued in OfflineSyncService and synced automatically when network returns.
- **Dynamic Server IP Configuration:** Dedicated in-app dialog allowing physical mobile devices to connect to local development IPs.
- **4-Second HTTP Timeout:** Prevents infinite network hanging on physical devices, falling back gracefully to cached datasets.

2.7 Lock-Screen Study Mode (`LockscreenService`)

An innovative background Android service that automatically prompts a rapid flashcard whenever the student wakes or unlocks their phone, converting passive screen unlocks into high-retention study moments.

3. Technical Architecture & Dependency Injection

Layer / Service	Provider Identifier	Architectural Responsibility
Presentation Layer	tiktokFeedProvider languageProvider authProvider	Riverpod StateNotifiers managing reactive UI state, theme, directionality, and localized viewports.
Domain & Repository	flashcardRepositoryProvider	Orchestrates offline caching, network fetches, priority deck shuffling, and offline review synchronization.
Cache Engine	cacheManagerProvider offlineSyncServiceProvider	Manages serialized JSON cache with TTL invalidation and persistent review synchronization queues.
Network & Storage	apiClientProvider storageServiceProvider	Handles HTTP communication with 4s timeouts, auth headers, and persistent SharedPreferences storage.

4. Quality Assurance & System Verification

The codebase adheres to rigorous clean code standards, validated through static analysis and version-controlled Git synchronization:

- **Dart Static Analysis:** dart analyze lib completed with **0 errors and 0 warnings**.
- **Backend Repository:** <https://github.com/qcmlab/vixa-backend.git> (FastAPI, SM-2, JWT Auth).
- **Admin Dashboard:** <https://github.com/qcmlab/vixa-admin.git> (React, Tailwind, Multilingual).
- **Mobile Repository:** <https://github.com/qcmlab/vixa-mobile.git> (Flutter, Riverpod, Offline-first).